BudgetBee – Small savings, bigger goals

Smart Personal Finance and Budgeting System

Using Data Structures and Algorithms

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B.Tech CSE (AI & ML) – Semester III

DSA with C/C++

Team Details

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Project Information

Project Title	BudgetBee – Small savings, bigger goals
Subject	DSA with C/C++
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A simple DSA-based approach to understand, control and improve personal spending.

1. Proposal Description

1.1 Motivation

Managing personal expenses becomes difficult when payments come from different sources and at different times. Small daily expenses can easily go unnoticed, while fixed payments such as rent, EMIs, insurance and subscriptions continue every month.

BudgetBee is proposed as a simple C++ based personal finance tracker that helps a user record expenses, organize them into categories and understand where money is being spent.

The main idea is to keep the system easy to use while applying the data structures and algorithms learned in DSA.

1.2 State of the Art / Current Solution

Many existing finance applications provide expense tracking and budgeting features. However, they often contain many options that may not be necessary for a student or an individual who simply wants to understand monthly spending.

BudgetBee focuses on the core problem: recording expenses, finding useful information quickly and giving simple suggestions based on the stored data.

The project also gives practical use to concepts such as linked lists, hashing, sorting, searching and trees.

1.3 Project Goals

The main goals of BudgetBee are:

- Record income and daily expenses.
- Organize transactions into useful categories.
- Track fixed payments such as EMI, rent, insurance and subscriptions.
- Search transactions quickly.
- Sort expenses according to amount, date or category.
- Set monthly budgets and savings goals.
- Show a simple safe-to-spend amount.
- Identify unusual spending patterns.
- Provide a basic what-if simulation before making a large expense.

1.4 Scope

The project is designed as a standalone C++ application developed in Visual Studio Code. It is intended mainly for personal use and academic demonstration.

The system will focus on data storage, processing, searching, sorting and basic financial analysis rather than online banking or real-time payment processing.

2. Milestones and Project Approach

The development of BudgetBee will be divided into small stages so that each part can be tested before moving to the next one.

Phase	Work	Expected Result
1	Requirement Analysis	Identify common personal expenses, user needs and important project features.
2	Data Structure Design	Select suitable structures for storing, searching and organizing transactions.
3	Core Development	Implement transaction entry, income/expense tracking and category handling.
4	Budget Features	Add monthly budgets, savings goals, EMI and recurring payment tracking.
5	Smart Analysis	Implement safe-to-spend calculation, unusual expense detection and what-if analysis.
6	Testing and Refinement	Test different inputs, edge cases and overall program flow.

2.1 Project Approach

The project will follow a modular approach. Each feature will be implemented separately and then connected through the main program.

First, transaction data will be collected from the user. The data will then be stored using suitable structures. Searching and sorting algorithms will help the user find required records. Finally, the processed information will be used to generate budgets, summaries and simple financial suggestions.

2.2 Real-Life Expense Categories

BudgetBee will support practical categories such as:

Category	Examples	Type
Housing	Rent, maintenance	Fixed
Food	Groceries, eating out	Variable
Utilities	Electricity, water, internet	Regular
Transport	Fuel, bus, cab	Variable
EMI / Loans	Education, vehicle, personal loan	Fixed
Subscriptions	OTT, music, software	Recurring
Insurance	Health, vehicle, life	Regular
Shopping	Clothes, electronics	Variable
Education	Books, courses, stationery	Variable
Other	Miscellaneous expenses	Variable

2.3 Development Environment

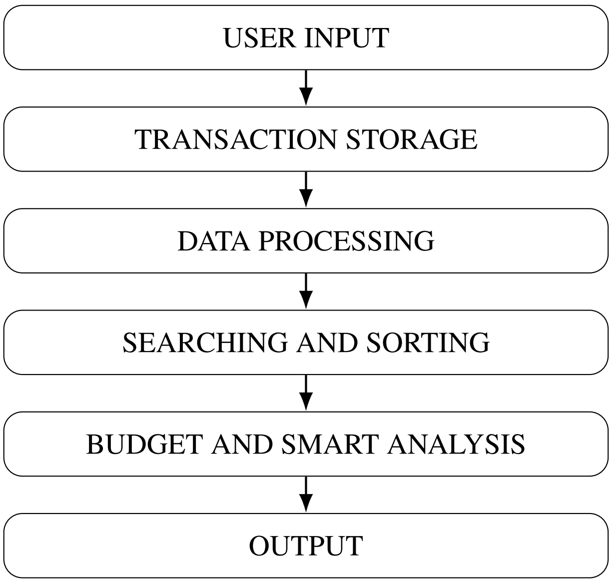
- Programming Language: C++
- IDE / Editor: Visual Studio Code

- Subject Concepts: Data Structures and Algorithms
- Platform: Standalone computer-based application

3. System Architecture and Outcome

3.1 Basic System Flow

The overall working of BudgetBee can be represented using the following simple flow:



3.2 Project Outcome

At the end of the project, BudgetBee will provide a working C++ application where a user can:

- Add and view income and expenses.
- Organize transactions into categories.
- Search for particular transactions.
- Sort expenses for easier comparison.
- Track monthly budgets.
- Monitor EMIs and recurring payments.
- Set savings targets.
- Get a simple spending summary.
- Check how a new expense may affect the remaining budget.

3.3 Expected Working

When the program starts, the user will be able to select the required operation from the menu. Transaction details such as amount, date, category and description will be entered and stored.

The program will process the stored information whenever the user requests a search, sorting operation, budget calculation or summary. The result will then be displayed in a simple text-based format.

The focus is on making the program understandable while demonstrating practical use of DSA concepts.

3.4 Academic Relevance

BudgetBee provides a practical problem for applying DSA concepts. Instead of using data structures only for theoretical examples, the project connects them with a real-life problem.

The implementation will demonstrate how the choice of a data structure can affect searching, insertion, organization and retrieval of financial records.

4. Unique Features and Assumptions

BudgetBee includes several features that make it more useful than a basic expense list.

4.1 Safe-to-Spend Amount

The program calculates an approximate amount that can still be spent after considering income, essential expenses, upcoming fixed payments and the user's savings target.

This gives the user a quick idea of how much money is reasonably available.

4.2 Bee Brain

The "Bee Brain" feature gives short suggestions based on spending patterns.

For example, if spending in a particular category increases significantly, the program can show a simple message suggesting that the user review that category.

The suggestions are rule-based and are intended to demonstrate basic logical analysis rather than machine learning.

4.3 EMI Tracker

The user can store details of an EMI such as amount, due date and remaining payments.

This helps separate loan-related expenses from normal monthly spending.

4.4 Recurring Payment Tracker

Regular payments such as subscriptions, insurance premiums and other recurring bills can be stored separately.

The feature helps the user remember payments that may otherwise be overlooked.

4.5 Unusual Expense Detection

BudgetBee can compare a new expense with previous spending in the same category.

If an expense is much higher than the user's usual amount, the program can flag it for review.

4.6 What-If Simulator

Before making a large purchase, the user can enter the expected amount.

The program can estimate how the purchase would change the remaining budget and savings position.

4.7 Savings Goals

The user can define a target such as buying a laptop, building an emergency fund or saving for a trip.

The system can show the progress towards the target based on the amount saved.

4.8 Assumptions

- The user enters correct transaction information.
- The project does not connect directly to bank accounts.
- Financial suggestions are basic rule-based suggestions.
- The application is intended for personal budgeting and academic demonstration.

- The system does not replace professional financial advice.

5. Implementation Details

5.1 Data Structures Used

The following DSA concepts are planned for the implementation:

Data Structure / Algorithm	Purpose	Application in BudgetBee
Linked List	Dynamic storage	Store transaction records as they are added.
Hash Map	Fast lookup	Find transactions or categories using a key.
BST	Ordered storage	Organize selected records for efficient searching.
Sorting	Ordering data	Arrange transactions by amount, date or category.
Searching	Finding records	Locate transactions according to user requirements.

5.2 System Working Steps

1. Start the BudgetBee application.
2. Select an option from the main menu.
3. Enter income or expense details.
4. Store the transaction using the selected data structure.
5. Apply searching or sorting when requested.
6. Calculate budget and savings information.
7. Check smart analysis conditions.
8. Display the required result.

5.3 Main Modules

The planned modules are:

- Transaction Module
- Category Module
- Budget Module
- EMI and Recurring Payment Module
- Search and Sort Module
- Savings Goal Module
- Smart Analysis Module
- Report / Summary Module

5.4 Testing Plan

Testing will be performed using different types of inputs, including:

- Normal income and expense entries.
- Zero and large transaction amounts.

- Multiple transactions in the same category.
- Empty transaction lists.
- Searches for existing and non-existing records.
- Budget limits that are reached or exceeded.
- Multiple recurring payments and EMIs.

The main objective of testing is to make sure that calculations, searches, sorting and menu operations produce the expected results.

6. References

The project will use the following resources for understanding and implementing the required concepts:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, *Introduction to Algorithms*, MIT Press.
2. Mark Allen Weiss, *Data Structures and Algorithm Analysis in C++*.
3. Bjarne Stroustrup, *The C++ Programming Language*.
4. C++ Reference, `https://en.cppreference.com/`
5. Microsoft Visual Studio Code Documentation, `https://code.visualstudio.com/docs`

7. Development Summary

BudgetBee is planned as a practical DSA project that connects personal finance with core programming concepts.

The project starts with simple transaction recording and gradually adds budgeting, searching, sorting, recurring payment tracking and savings-related features. The smart features are designed to remain simple enough to implement in C++ while still making the project useful in a real-life situation.

The main learning outcome is to understand how data structures and algorithms can be selected according to the problem instead of using them only as theoretical concepts.

Team Responsibilities

Member	Section	Primary Responsibility
Krishna Agarwal	ML4	Team coordination, core implementation and integration.
Disha Tahiliani	DS2	Data structure implementation and transaction handling.
Divyanshu Mehandiratta	DS1	Searching, sorting and data processing logic.
Abhigyan Rathi	ML3	Budget analysis, smart features and testing support.